

Written by ChatGPT

The Constant 11/15

A rigorous density derivation and its sieve, Fourier, and ODE interpretations

$$\frac{11}{15} = 0.73333333\ldots$$

Contents

1	The arithmetic set and its natural density	1
1.1	Rigorous proof by floor functions and inclusion–exclusion	1
1.2	Equivalent proof from periodicity modulo 30	2
2	Sieve and analytic-number-theory interpretation	3
3	Discrete Fourier interpretation	3
4	ODE analogue: classification and precise averaging	4
4.1	What type of ODE is this?	4
4.2	Rigorous comparison with the averaged autonomous equation	4
5	Relevant surrounding theory	5
6	When 11/15 is a strong candidate	5
7	References	6
	Appendix: Prompt History	7

1 The arithmetic set and its natural density

Define

$$S := \{n \in \mathbb{N} : 2 \nmid n \text{ or } 3 \nmid n \text{ or } 5 \nmid n\}.$$

Thus S is the set of positive integers removed by trial division by the primes 2, 3, 5.

For any $A \subseteq \mathbb{N}$, its *natural density*, when the limit exists, is

$$\delta(A) := \lim_{N \rightarrow \infty} \frac{1}{N} \#(A \cap \{1, \dots, N\}).$$

The symbol $\#$ denotes cardinality. The definition does not assume that the limit exists; existence must be proved.

1.1 Rigorous proof by floor functions and inclusion–exclusion

For $d \geq 1$, let

$$A_d := \{n \in \mathbb{N} : d \mid n\}.$$

For every integer $N \geq 1$,

$$\#(A_d \cap [1, N]) = \left\lfloor \frac{N}{d} \right\rfloor.$$

Since $S = A_2 \cup A_3 \cup A_5$, finite inclusion–exclusion gives the exact identity

$$\begin{aligned} C(N) := \#(S \cap [1, N]) &= \left\lfloor \frac{N}{2} \right\rfloor + \left\lfloor \frac{N}{3} \right\rfloor + \left\lfloor \frac{N}{5} \right\rfloor \\ &\quad - \left\lfloor \frac{N}{6} \right\rfloor - \left\lfloor \frac{N}{10} \right\rfloor - \left\lfloor \frac{N}{15} \right\rfloor + \left\lfloor \frac{N}{30} \right\rfloor. \end{aligned}$$

The denominators 6, 10, 15, 30 occur because intersections of divisibility conditions are governed by least common multiples; for example,

$$A_2 \cap A_3 = A_{\text{lcm}(2,3)} = A_6.$$

Write

$$\left\lfloor \frac{N}{d} \right\rfloor = \frac{N}{d} + \theta_d(N), \quad -1 < \theta_d(N) \leq 0.$$

Substitution into the exact counting formula yields

$$C(N) = N \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{5} - \frac{1}{6} - \frac{1}{10} - \frac{1}{15} + \frac{1}{30} \right) + E(N),$$

where

$$E(N) = \theta_2 + \theta_3 + \theta_5 - \theta_6 - \theta_{10} - \theta_{15} + \theta_{30}$$

and therefore $|E(N)| < 7$. The rational coefficient is

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{5} - \frac{1}{6} - \frac{1}{10} - \frac{1}{15} + \frac{1}{30} = \frac{15 + 10 + 6 - 5 - 3 - 2 + 1}{30} = \frac{22}{30} = \frac{11}{15}.$$

Consequently,

$$\frac{C(N)}{N} = \frac{11}{15} + \frac{E(N)}{N}, \quad \left| \frac{E(N)}{N} \right| < \frac{7}{N} \rightarrow 0.$$

This proves, rather than assumes, that the natural density exists:

$$\boxed{\delta(S) = \frac{11}{15}.}$$

1.2 Equivalent proof from periodicity modulo 30

Membership in S depends only on the residue class modulo 30. Exactly

$$\varphi(30) = 30 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 8$$

residue classes are coprime to 30, so the other $30 - 8 = 22$ classes lie in S .

More explicitly, write $N = 30q + r$ with $0 \leq r < 30$. Every complete block of 30 consecutive integers contributes exactly 22 elements of S , hence

$$C(N) = 22q + C(r).$$

Since $C(r)$ and r are bounded independently of q ,

$$\frac{C(N)}{N} = \frac{22q + C(r)}{30q + r} \rightarrow \frac{22}{30} = \frac{11}{15}.$$

This makes the existence of the limit especially transparent.

2 Sieve and analytic-number-theory interpretation

The complementary set consists of integers coprime to 30. Its density is

$$\frac{\varphi(30)}{30} = \prod_{p|30} \left(1 - \frac{1}{p}\right) = \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = \frac{4}{15}.$$

Therefore

$$\boxed{\frac{11}{15} = 1 - \frac{\varphi(30)}{30}} = 1 - \prod_{p|30} \left(1 - \frac{1}{p}\right).$$

In sieve terminology, 4/15 is the proportion surviving removal of multiples of 2, 3, 5, while 11/15 is the removed proportion. This is the first nontrivial “wheel” based on the modulus

$$2 \cdot 3 \cdot 5 = 30.$$

The same local factors $1 - 1/p$ are basic components of sieve estimates and Euler products. The value 11/15 is canonical when the structure genuinely singles out the primes 2, 3, 5 or period 30; it is not a universal limiting density as more primes are added.

3 Discrete Fourier interpretation

Define the arithmetic mask on the cyclic group $\mathbb{Z}/30\mathbb{Z}$ by

$$h(a) := \mathbf{1}_{\gcd(a,30)>1}.$$

Using the normalized discrete Fourier transform

$$\hat{h}(k) := \frac{1}{30} \sum_{a \bmod 30} h(a) e^{-2\pi i k a / 30},$$

the zero-frequency coefficient is its mean:

$$\boxed{\hat{h}(0) = \frac{1}{30} \sum_{a \bmod 30} h(a) = \frac{22}{30} = \frac{11}{15}}.$$

Thus 11/15 is exactly the DC component of the 2–3–5 divisibility mask.

For $k \neq 0$, define the Ramanujan sum

$$c_{30}(k) := \sum_{\substack{a \bmod 30 \\ \gcd(a,30)=1}} e^{-2\pi i k a / 30}.$$

Since the sum of a nontrivial character over all residues is zero and $h = 1 - \mathbf{1}_{(a,30)=1}$,

$$\hat{h}(k) = -\frac{c_{30}(k)}{30}, \quad k \neq 0.$$

The nonzero modes record the periodic arithmetic oscillation, while 11/15 records the mean. Any circulant convolution operator built from this mask is diagonalized by these Fourier characters.

4 ODE analogue: classification and precise averaging

Extend the mask to a one-periodic step function $f : \mathbb{R} \rightarrow \{0, 1\}$ by assigning one interval of length $1/30$ to each residue class modulo 30 and setting the value equal to h on that interval. Then

$$m := \int_0^1 f(\tau) d\tau = \frac{22}{30} = \frac{11}{15}.$$

Consider, for fixed $\varepsilon > 0$,

$$y'_\varepsilon(t) = -\lambda f(t/\varepsilon) y_\varepsilon(t), \quad y_\varepsilon(0) = y_0.$$

4.1 What type of ODE is this?

For each fixed $\varepsilon > 0$, it is a scalar, first-order, linear, homogeneous, *non-autonomous* ODE with a rapidly periodic, piecewise-constant coefficient. Because the coefficient is bounded and measurable, the natural solution is absolutely continuous and satisfies the equation almost everywhere. It is given exactly by

$$y_\varepsilon(t) = y_0 \exp \left(-\lambda \int_0^t f(s/\varepsilon) ds \right).$$

It is not autonomous as a one-dimensional equation. Introducing the phase variable $\theta \in \mathbb{R}/\mathbb{Z}$ gives the autonomous lifted system

$$\theta' = \frac{1}{\varepsilon}, \quad y' = -\lambda f(\theta)y,$$

on the cylinder $(\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$.

4.2 Rigorous comparison with the averaged autonomous equation

Set $g = f - m$. Then g is one-periodic and has zero mean. Define

$$G(u) := \int_0^u g(v) dv.$$

Because $\int_0^1 g = 0$, one has $G(u+1) = G(u)$; hence G is bounded. A change of variables gives the exact identity

$$\begin{aligned} \int_0^t f(s/\varepsilon) ds &= \varepsilon \int_0^{t/\varepsilon} f(u) du \\ &= mt + \varepsilon G(t/\varepsilon). \end{aligned}$$

Therefore

$$y_\varepsilon(t) = y_0 e^{-\lambda m t} e^{-\lambda \varepsilon G(t/\varepsilon)}.$$

The averaged autonomous equation is

$$\bar{y}'(t) = -\lambda m \bar{y}(t) = -\frac{11\lambda}{15} \bar{y}(t), \quad \bar{y}(0) = y_0,$$

with solution $\bar{y}(t) = y_0 e^{-\lambda m t}$. If $M := \sup_{u \in \mathbb{R}} |G(u)| < \infty$, then for every fixed $T > 0$,

$$\sup_{0 \leq t \leq T} \left| \frac{y_\varepsilon(t)}{\bar{y}(t)} - 1 \right| \leq e^{|\lambda| \varepsilon M} - 1 = O(\varepsilon).$$

Thus $y_\varepsilon \rightarrow \bar{y}$ uniformly on bounded time intervals. This is the precise sense in which the non-autonomous equation is close to an autonomous ODE when ε is small.

There is an additional exact Floquet statement. Over any one fast period,

$$\int_t^{t+\varepsilon} f(s/\varepsilon) \, ds = \varepsilon \int_{t/\varepsilon}^{t/\varepsilon+1} f(u) \, du = \varepsilon m.$$

Hence the monodromy multiplier is exactly

$$\frac{y_\varepsilon(t+\varepsilon)}{y_\varepsilon(t)} = e^{-\lambda \varepsilon m} = e^{-11\lambda \varepsilon/15},$$

and its Floquet exponent per unit time is exactly

$$\boxed{-\lambda m = -\frac{11\lambda}{15}.$$

So in this scalar multiplicative model, $11\lambda/15$ is not merely an informal leading-order rate: it is the exact long-time exponential decay rate when $\lambda > 0$.

5 Relevant surrounding theory

- **Elementary and combinatorial number theory:** inclusion–exclusion, residue classes, least common multiples, and Euler’s totient.
- **Analytic number theory and sieve methods:** finite Euler products, pre-sieving, wheel sieves, and local congruence densities.
- **Discrete Fourier analysis:** Fourier coefficients of periodic arithmetic masks, Ramanujan sums, and circulant operators.
- **ODEs and Floquet theory:** scalar linear equations with periodic coefficients, monodromy, and characteristic exponents.
- **Averaging and homogenization:** rapidly oscillating coefficients replaced by their period mean, with an explicit $O(\varepsilon)$ error here.
- **Discrete analogues:** multiplicative recurrences whose logarithmic growth or decay is governed by the same mask average.

6 When 11/15 is a strong candidate

The identification is structurally persuasive when the underlying object contains at least one of the following:

- modulus or period 30;
- the primes 2, 3, 5 or their product;
- exclusion of multiples of 2, 3, 5;

- a binary mask with exactly 22 active positions in each block of 30;
- an averaged coefficient or zero Fourier mode of such a mask;
- a wheel sieve or local congruence obstruction.

Numerical proximity without one of these signatures is weaker evidence.

7 References

1. T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976.
2. H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS, 2004.
3. H. Montgomery and R. Vaughan, *Multiplicative Number Theory I*, Cambridge University Press, 2007.
4. NIST Digital Library of Mathematical Functions, Section 27.10, “Periodic Number-Theoretic Functions” (Ramanujan sums).
5. E. A. Coddington and N. Levinson, *Theory of Ordinary Differential Equations*, McGraw–Hill, 1955.
6. J. A. Sanders, F. Verhulst, and J. Murdock, *Averaging Methods in Nonlinear Dynamical Systems*, 2nd ed., Springer, 2007.

Appendix: Prompt History

The following are the user's prompts preceding the PDF-revision requests. They are reproduced verbatim except for typographic line wrapping.

1. "0.730-0.740
Are there any interesting mathematical constants which are found in this bound?"
2. "0.733-0.734 In particular?"
3. "I am working with anything that might be related to ODE theory, analytic number theory, fourier analysis, in any form."
4. "I am working with anything that might be related to ODE theory, analytic number theory, fourier analysis, in any form, including discrete analogues, prime number properties."
5. "a_65, a_205 is this converging to some number that we haven't yet fully identified? Whats the surrounding theory, it seems like a possible candidate.

Whats the motivation for studying that, any relevant connections to open conjectures?

Keep responses brief, jsut exploring."
6. "Okay, it must be specifically within the bounds I specified uncondtionally."
7. "Expand upon the 11/15, 0.73333333 candidate, it seems about the closest to me"
8. "That is an extremely compelling possibility, for my use-case, please prepare a pdf deriving that figure and brief explanation of relevant theory areas."